# Maiqi Jiang

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## Education

**PhD** **College of William & Mary**, Computer Science — Williamsburg, VA
- Advisor: Prof. Yanfu Zhang — Aug 2025 – present

**M.Sc. (with Distinction Award)** **The Hong Kong Polytechnic University (QS Top 60)**, Information Technology — Hong Kong, Aug 2021 – Jan 2023

**B.S.** **Northeastern University, China (985 Project)**, Biomedical Engineering — Shenyang, China, Sept 2017 – June 2021

## Experience

**William & Mary**, PhD Researcher — Federated Tuning & Structural Pruning for LLMs (SFT-P) — Williamsburg, VA, Aug 2025 – Jan 2026, 6 months
- Advisor: Prof. Yanfu Zhang
- Proposed **SFT-P**: FedAvg-based training that jointly learns client-specific structured pruning masks and adaptation for on-device deployment under non-IID private data; supports mixed client budgets.
- Implemented a client-conditioned mask generator (shared hypernetwork + private client embedding) producing hard binary routing masks via Bernoulli gating, and a residual-safe route/merge interface for dense-kernel execution; achieved **+8.5 Avg over the best federated baseline on LLaMA-7B at 50% pruning**. Submitted to **NAACL 2027**.

**William & Mary**, PhD Researcher — In-Context Knowledge Editing for LLMs (DR-IKE / FWE-IKE / MO-IKE) — Williamsburg, VA, Aug 2025 – present, 1 year 3 months
- Advisors: Prof. Yanfu Zhang, Prof. Haipeng Chen
- Main contributor to a research line on **retrieval-based in-context knowledge editing** for black-box LLMs, spanning policy-optimized retrieval, confidence-aware fusion, and constrained multi-objective RL.
- **DR-IKE** (EMNLP 2025, co-first): trained a BERT retriever via REINFORCE with a learnable threshold to dynamically size demonstrations. [code]
- **FWE-IKE** (ACL 2026 Findings): confidence-gated ensemble of LLM/BERT/MLP rankers with logit-space fusion, closing ~53% of the gap to the oracle on COUNTERFACT.
- **MO-IKE** (EMNLP 2026 Findings): formulated prompt construction as a Constrained MDP with Lagrangian relaxation and GRPO, jointly optimizing reliability, generality, and specificity.

**William & Mary**, PhD Researcher — Meta-Path Attention Reliability in Heterogeneous GNNs (MetaXplain) — Williamsburg, VA, Aug 2025 – Feb 2026, 7 months
- Advisor: Prof. Yanfu Zhang
- Conducted a controlled empirical study of whether meta-path attention is a **faithful explanation**, identifying regimes of alignment vs. statistically significant decoupling across multiple datasets and HGNN backbones.
- Introduced **MetaXplain**, a meta-path-aware post-hoc analysis protocol (view-factorized explanations, schema-valid perturbations, fusion-aware attribution), and proposed **MP-AEA** to quantify attention–explanation alignment; released an explainability benchmark codebase ExplainerForHGNN for the community. Submitted to **TMLR** (under review).

**William & Mary**, PhD Researcher — Biomedical HIN Link Prediction (IM-HIN / PAIRS) — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Aug 2025 – Feb 2026
- Proposed **PAIRS**: framed protein-phenotype association prediction as **link prediction** over a biomedical heterogeneous information network, rather than per-label classification. — 7 months
- Built **IM-HIN** by integrating STRING, HumanNet XN, GeneMania (protein-protein interactions) and HPO (protein-phenotype associations) into a single graph; two type-specific encoders (protein encoder + phenotype encoder) feed a relation-prediction head.
- Accepted at **ICASSP 2026** (first author).

**William & Mary**, Project Lead — In-Context Task Learning: Reading Out a Kernel Classifier — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Feb 2026 – present
- Identified **mapping-sensitive attention heads** through paired intact/randomized-label contrasts and characterized their outputs as a **kernel classifier** combining query-dependent retrieval with contextual class votes. On **TREC-fine** with Llama-3.1-8B and 180 demonstrations, direct readout from **8 selected heads** achieves **50.3% accuracy**, versus 51.6% for the full model. — 9 months
- Proposed **Selective TL Memory**, reusing cached demonstration K/V through selected heads with live query-dependent retrieval. Selective access improves the same fixed readout across **all 18 natural-language settings** at both tested head budgets; the **30-head readout** outperforms all compared adapted-vector baselines in **14/18 settings**. Submitted to **ICLR 2027** (under review).

**William & Mary**, Project Mentor — DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Dec 2025 – Mar 2026
- Mentored two interns (the first two authors); guided the design of **DyV2X**, a cooperative trajectory prediction framework that constructs per-timestep dynamic interaction graphs and uses a temporal graph attention encoder to jointly model evolving inter-agent interactions and individual kinematic history. — 4 months
- On V2X-Seq-TFD (9,320 target agents), improved over the V2X-Graph baseline on all metrics: **ADE** 1.110→0.952 (**14.2%↓**), **FDE** 1.894→1.664 (**12.1%↓**), **MR@2m** 27.9%→24.7%.
- Largest gains on safety-critical turning maneuvers (e.g., *left turn*: FDE 4.216→2.646, **37.2%↓** MR@2m 64.2%→47.5%). Submitted to **ICRA 2027** (under review).

**William & Mary**, Project Mentor — Intern-led Projects on LLM Efficiency & Long-Context — Williamsburg, VA

- Advisor: Prof. Yanfu Zhang — Sept 2025 – present
- **Domain-Adaptive Pruning (CDAM).** Mentoring *Noman Ali* on a **training-free activation measure** combining domain-specific and general-purpose statistics with feature-wise cross-domain interactions, integrated into Wanda and CFSP. On Llama-2-13B at **50% sparsity** with 1,024 calibration examples, CDAM raises CFSP's average biomedical accuracy from **34.45% to 38.70%** versus naive calibration mixing, while improving general-task accuracy from 58.73% to 61.32%. Targeting **ACL 2027**. — 1 year 2 months
- **ICL-Preserving Pruning (ICLPrune).** Mentoring *Chongyuan Sun* on improving pruned-LLM **in-context learning** performance under task-shifted prompts via demonstration-aware calibration (*ICLPrune: Improving In-Context Learning in Structured LLM Pruning via Demonstration-Aware Calibration*, targeting **ACL 2027**).
- **Long-Context & ICL.** Proposed the research direction and co-supervised (with Prof. Zhang) *Yiran Ding* on the **cross-lingual gap in long-context reasoning** under context-extension methods, leading to *One Direction Away: The Cross-Lingual Gap in Long-Context Reasoning and a Shift That Closes It* (submitted to **NAACL 2027**); mentoring *Muhammed Muminul Hoque* on demonstration-ordering policies for ICL motivated by the **Lost-in-the-Middle** effect, given per-example rankings.

**Independent / Open Source**, Project Lead — lmdiff: Behavioral Diagnostics for LLM Configurations — Williamsburg, VA; Oct 2025 – present; 1 year 1 month

- Advisor: Prof. Yanfu Zhang
- Built and released **lmdiff** (PyPI `lmdiff-kit`, GitHub), a Python library for comparing LLM **configurations**—weights + system prompt + decoding + adapter—via behavioral distance and multi-level diagnostics, rather than aggregate benchmark scalars.
- Centered the API on a `family()` abstraction that compares one base configuration against N variants on a shared set of probe-domain axes, treating weight modifications (fine-tuning, RLHF, adapters), runtime modifications (system prompt, decoding, ICL context), and quantization/pruning as first-class variants. Geometry-level metrics (per-domain drift, share, raw + selective cross-variant cosine) quantify *where* each variant moves and whether different modifications push in the same direction. Showcase 7-variant Llama-2-7B family reveals a system prompt can reshape next-token distributions more than 7B→13B scaling.
- Engineering: engine layer with HF Transformers + mock backends, engine reuse that keeps peak VRAM at base + 1 active variant in 7-variant families, lm-eval-harness multi-task probe loader with task→domain mapping, JSON schema versioning with auto-recompute on load, 5-channel reporting (ANSI / PNG / HTML / Markdown / JSON), ~860 tests. Active development; representation, trajectory, and causal metrics planned for v0.4+.

**The Hong Kong Polytechnic University**, Researcher — Automated Heterogeneous Network Learning (AutoGNR) — Hong Kong; Nov 2021 – Feb 2023; 1 year 4 months

- Advisor: Prof. Xiao Huang
- Co-developed **AutoGNR**, using differentiable NAS to automatically select heterogeneous aggregation paths and node-type combinations—conceptually aligned with structured pruning and bi-level optimization. [AutoGNR] [HINBaselines]
- Designed a non-recursive message-passing scheme that decouples path search from depth, improving search efficiency on large heterogeneous graphs.

**City University of Hong Kong**, Research Assistant — Temporal User Engagement Modeling in Polarized Social Platforms — Hong Kong; Feb 2023 – Dec 2023; 11 months

- Advisor: Prof. Junming Liu
- Fused image/video content features with text and interaction context via a temporal GNN for forecasting user engagement across polarized communities on Twitter/Reddit.
- Built multimodal feature pipelines and longitudinal datasets for studying engagement dynamics under polarization. [RedditProcess] [RedditDataset]
- Submitted as a short paper to **ICIS 2023**, proposing a counter-party graph attention network with sentiment diffusion for online engagement prediction.

**Neusoft Inc.**, Software Developer Intern — Shenyang, China; Sept 2020 – Sept 2020; 1 month

**Northeastern University, China**, Undergraduate Researcher / Bachelor's Thesis — Semi-supervised Learning for Whole-Slide Pathology Images (WSI) — Shenyang, China; Nov 2018 – June 2021; 2 years 8 months

- Advisor: Prof. Xiaoyu Cui
- Developed weakly-/semi-supervised pipelines for gigapixel WSI using eye-tracking signals, focusing on efficient patch sampling and large-scale image processing. [Modeling] [Processing]
- Maintained reusable WSI tooling (tiling, feature extraction, dataset management) used across the lab's projects.

## Publications

**PAIRS, Not Labels: Predicting Protein-Phenotype Associations via Link Prediction** — Apr 2026
Maiqi Jiang, Yanshuo Chen, Guodong Liu, Avinash Sahu, Ye Gao, Yanfu Zhang

**Dynamic Retriever for In-Context Knowledge Editing via Policy Optimization (DR-IKE)** Nov 2025
Mahmud Wasif Nafee, *Maiqi Jiang*, Haipeng Chen, Yanfu Zhang
aclanthology.org/2025.emnlp-main.848

**Towards Reliable, Generalizable, and Specific In-Context Knowledge Editing via Multi-Objective Reinforcement Learning (MO-IKE)** Nov 2026
Xuzhong Wang, Maiqi Jiang, Tejal Nair, Girija Bhusal, Yanfu Zhang, Haipeng Chen

**Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing (FWE-IKE)** July 2026
Tejal Nair, Mahmud Wasif Nafee, Maiqi Jiang, Ashley Gao, Haipeng Chen, Yanfu Zhang

**Hierarchical Convolution Multibranch Transformer for EEG Signals** Apr 2026
Omid Mersa, Maiqi Jiang, Ye Gao, Qun Li, Yanfu Zhang

**In-Context Task Learning: Reading Out a Kernel Classifier** Sept 2026
Maiqi Jiang, Xugui Zhou, Bin Ren, Guodong Liu, Yanfu Zhang

**SFT-P: Federated Tuning and Pruning for LLMs** Oct 2026
Maiqi Jiang, Xugui Zhou, Bin Ren, Junyi Li, Shangqian Gao, Yanfu Zhang

**One Direction Away: The Cross-Lingual Gap in Long-Context Reasoning and a Shift That Closes It** Oct 2026
Yiran Ding, Maiqi Jiang, Qun Li, Yanfu Zhang

**Representation Change and Adaptive Stopping in In-Context Learning** Oct 2026
Tejal Nair, Omid Mersa, Zhen Xu, Maiqi Jiang, Xuzhong Wang, Ashley Gao, Gang Zhou, Haipeng Chen, Yanfu Zhang

**Is Meta-Path Attention an Explanation? Evidence of Alignment and Decoupling in Heterogeneous GNNs** Feb 2026
Maiqi Jiang, Noman Ali, Yiran Ding, Yanfu Zhang
openreview.net/forum?id=g9cnIFAYB3

**Disentangling Memorization in Language Models via Sparse Representation Intervention** Sept 2026
Jiarui Li, Maiqi Jiang, Junyi Li, Bin Wang, Shangqian Gao, Yanfu Zhang

**DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction** Sept 2026
Yiran Ding, Muhammed Muminul Hoque, Maiqi Jiang, Yongming Qin, Sidi Lu, Yanfu Zhang

**Automated Heterogeneous Network Learning with Non-Recursive Message Passing (AutoGNR)** Jan 2025
Zhaoqing Li, *Maiqi Jiang*, Shengyuan Chen, Bo Li, Guorong Chen, Xiao Huang
arxiv.org/abs/2501.07598

## Teaching

- TA — Topological Data Analysis (CSCI 520/420), William & Mary, Spring 2026
- TA — Neural Networks for Machine Learning (CSCI 446/546), William & Mary, Fall 2025

## Technical Skills

**LLM Training & Adaptation:** Federated LLM tuning with FedAvg; structured pruning with learned binary masks and dense-kernel route/merge execution (LLaMA-7B); HuggingFace + PyTorch.

**RL for LLMs:** REINFORCE-based and GRPO-based retrieval / policy optimization for in-context knowledge editing (DR-IKE, MO-IKE); reward design under Constrained MDP with Lagrangian relaxation.

**Graph Learning:** Heterogeneous GNNs (HAN/HGT-family), temporal graph networks (TGN / DyGFormer), differentiable NAS over heterogeneous aggregation paths; PyG-style implementation.

**Mechanistic Interpretability:** Attention head analysis for ICL; sparse autoencoder (SAE) representation interventions with ReFT; meta-path attention faithfulness diagnostics.

**Systems & Tooling:** Python library design and PyPI release (`lmdiff-kit`); multi-codebase open-source benchmarks (ExplainerForHGNN, HINBaselines); large-scale data pipelines for WSI and multimodal social media; mentoring code reviews across 4 intern projects.

**Languages:** Python, PyTorch, C++. Spoken: Chinese (native), English (fluent).

## Research Interests

- What interventions actually change inside an LLM — structured pruning, knowledge editing, adapters, decoding
- Building interventions that preserve the capabilities that matter (in-context learning, domain knowledge, factual reliability)
- Mechanistic analysis of in-context learning and of attention-based explanations
- RL-based methods for LLMs (policy-optimized retrieval, constrained multi-objective RL)
- Federated / on-device LLM adaptation and model compression
- Behavioral diagnostics and released tooling for LLM configurations
- Graph learning and explainability (heterogeneous GNNs, biomedical HINs, V2X)

## Service

- Research Assistant: City University of Hong Kong, with Prof. Junming Liu (Feb 2023 – Dec 2023)
- Software Developer Intern: Neusoft Inc., China (Sep 2020)
- PC member / Reviewer: **ICASSP 2026**, **NeurIPS 2026**, **CIKM 2026 Workshop**, **ICLR 2027**, **ACM Transactions on Computing for Healthcare**
- Author & maintainer of lmdiff-kit (PyPI)
- Open Source Contributor: The Unified AI Framework (Jul 2023); ExplainerForHGNN benchmark (2025–2026)